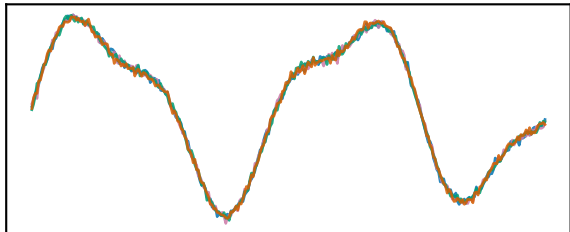


Spatial (graph) encoding of PM_{2.5} across IoT stations

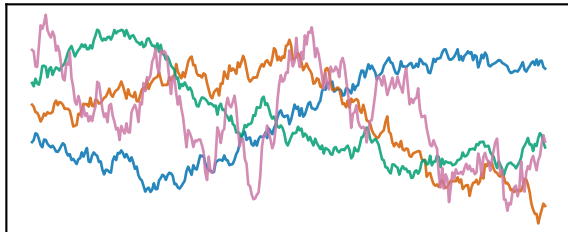
Homogeneous network
(e.g. Beijing / CZT, $\bar{r} \approx 0.88-0.90$)



time

neighbours nearly identical
⇒ aggregation redundant
⇒ small penalty

Heterogeneous network
(e.g. London / Madrid, $\bar{r} \approx 0.4-0.5$)



time

neighbours dissimilar
⇒ averaging mixes signals
⇒ large penalty